



Predicting Failure Before It Becomes Expensive



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Google for Startups

One Program Forced Us to Build a Platform

- Started as a therapeutics company
- Needed to spot risk earlier
- Built an internal AI prediction engine
- Realized it could scale far beyond one asset

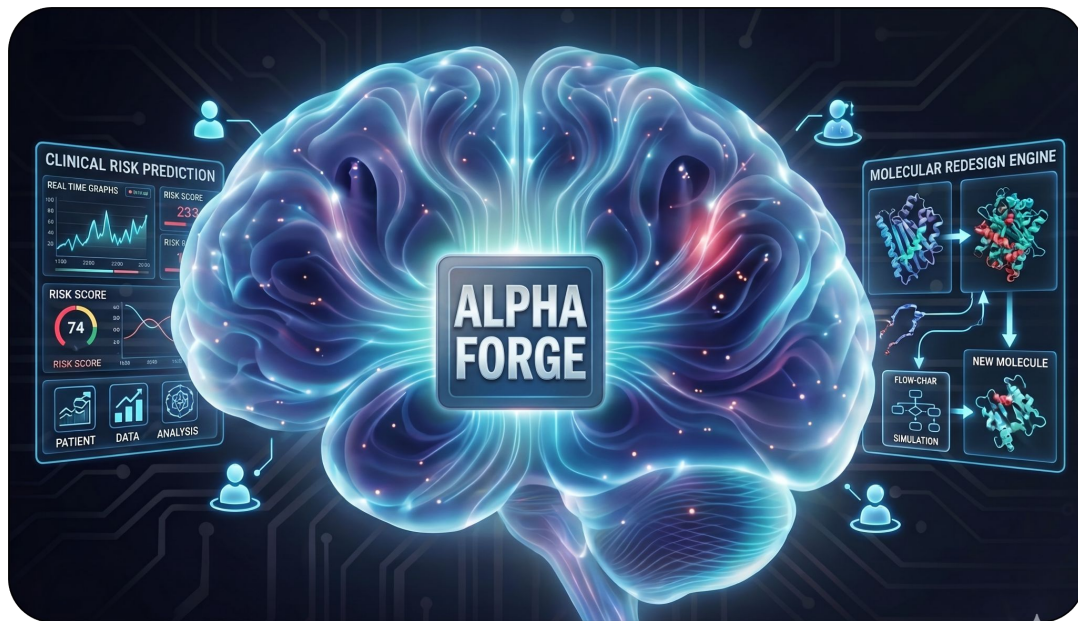


Drug Failure is a Design Failure

- Many failures are driven by risks encoded in molecules early
- Those risks are usually discovered too late
- The industry spends years & millions optimizing compounds that were never viable



Our AI Predicts Failure, Explains Why, & Redesigns Earlier



Score risk | Flag liabilities | Design lower-risk molecules

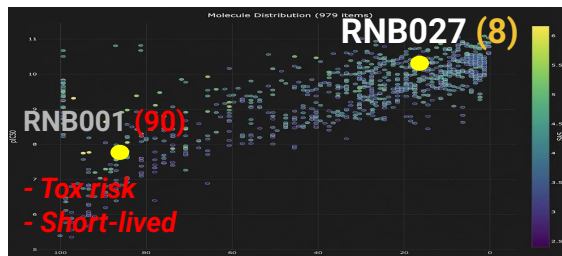
We Used AlphaForge to Rescue Our Own Program



Scored Internal program

- RNB001 (3rd Gen Academic Lead)
- Platform Score: **90** (High Risk)
- Drives regeneration & lowers inflammation
- Improves cognition

Flagged liabilities



Designed RNB027 (lower-risk)

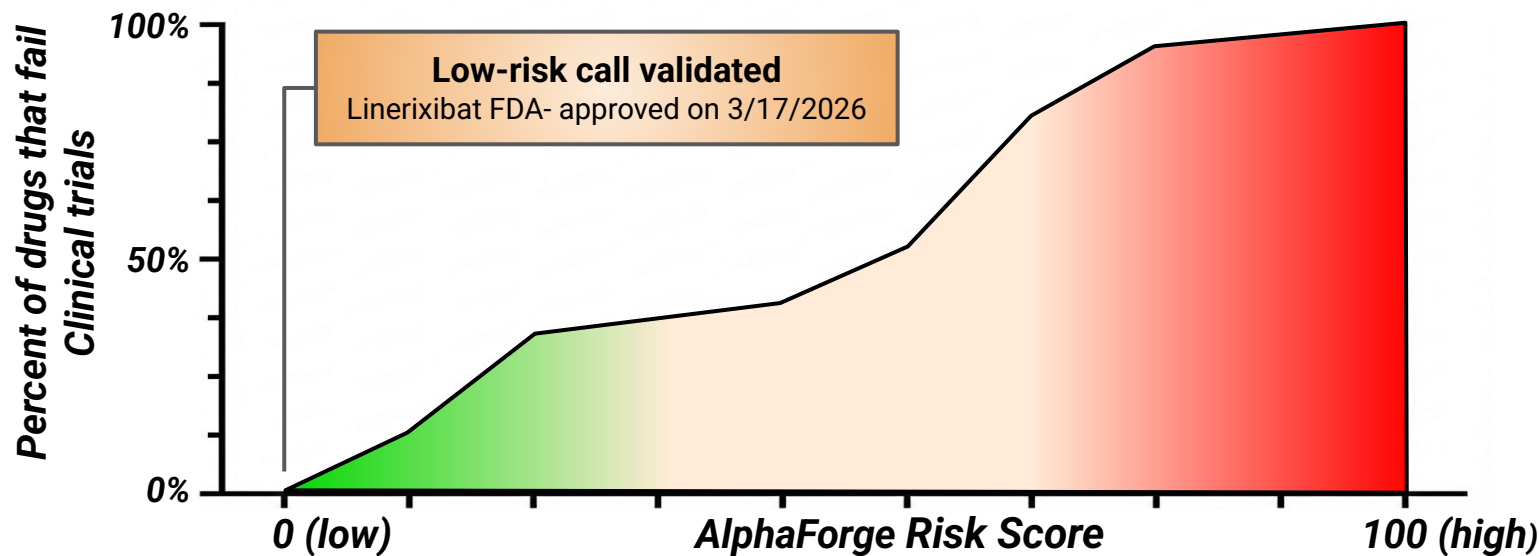
AlphaForge redesign

- Risk score: **90** → **8**
- Performance: **3x PD improvement**
- Speed: **<2 months**
- Cost: **< \$50K**

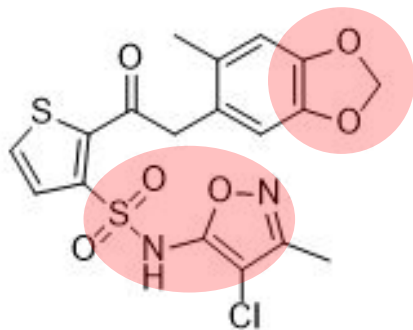
AlphaForge Risk Score Tracks Real Clinical Outcomes

Higher score = higher failure risk

393 compounds | outcomes not used for training | AUC 0.884



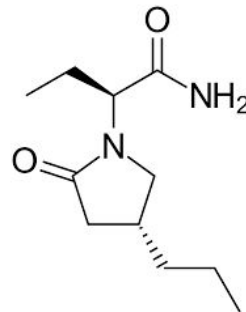
Auditable Assessment of Compounds



SITAXENTAN

Approved; later withdrawn due to severe liver toxicity risk

- ❑ Platform flags **structural liabilities**:
 - ❑ Benzodioxole → CYP-deactivation risk
 - ❑ Sulfonamide/isoxazole → hypersensitivity risk
 - ❑ Combined metabolic liability → elevated liver tox risk
- ❑ **Risk Score: 95 (High Risk)**

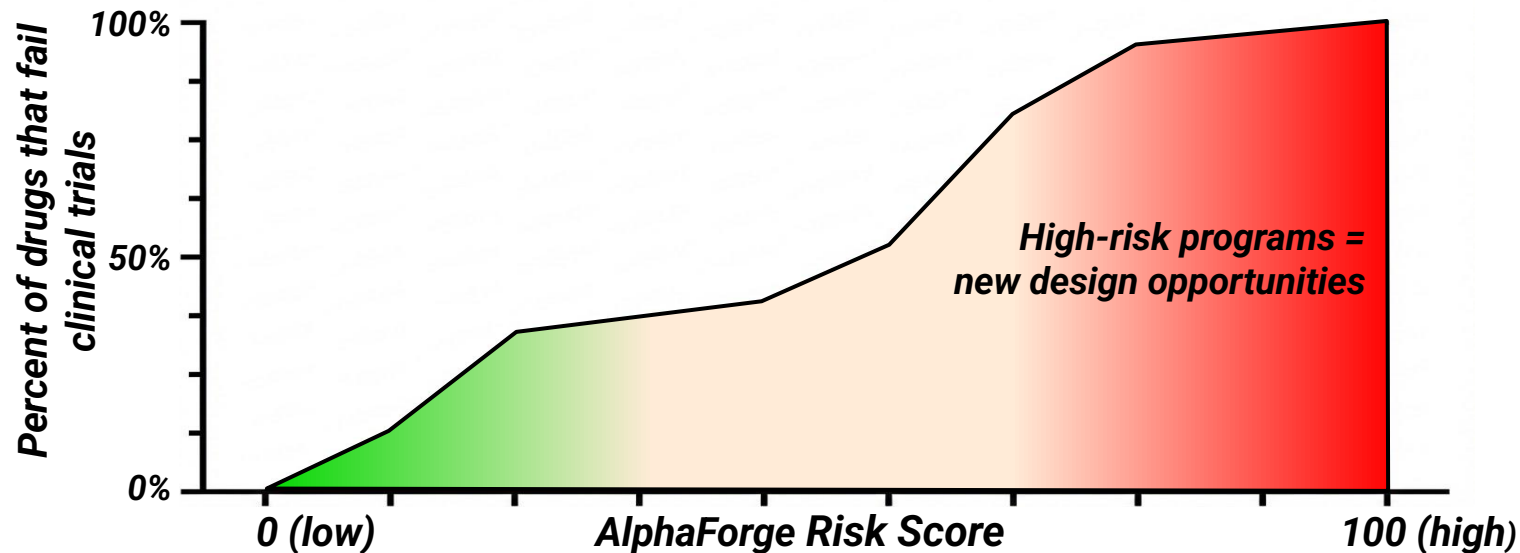


BRIVARACETAM

Approved CNS Drug: Brivaracetam

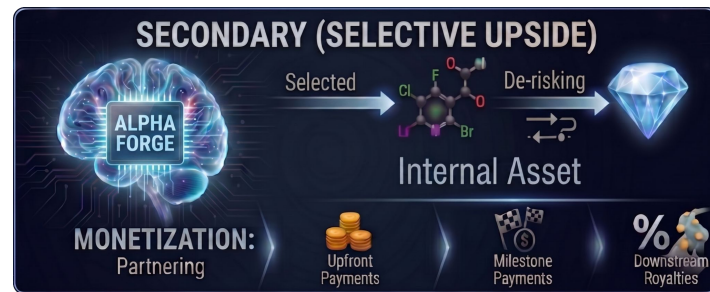
- ❑ Balance physicochemical properties
- ❑ No major structural liabilities identified
- ❑ **Risk Score: 5 (Low Risk)**

From Internal Proof to Pipeline Expansion



Ongoing collaboration with UC Berkeley's Molecular  **mti**
Therapeutics Initiative to apply our platform to their programs

Same Engine. First Prove Demand, Then Selectively Own Upside.



Near-term partner value. Long-term asset upside. Capital-efficient by design.

Built by Leaders in AI Drug Design, Medicinal Chemistry, & CNS Translation

Founding Team



Salah Mahmoudi, PhD
Co-founder, CEO | Drug discovery
& company-building



Tyson Ruetz, PhD
Co-founder, CSO | Aging brain
biology & target discovery



Stylianos Kyriacou, PhD
Co-founder, CTO | AI platform &
molecular design



Ramil Baiazitov, PhD
Head of Medicinal Chemistry | 19 years
in pharma; 3 assets to clinic, 1 approved



Selected Advisors/Collaborators



Anne Brunet, PhD
World-Leading Expert in Aging Biology |
Professor of Genetics | Stanford University



Julia Schaletzky, PhD
Executive Director, Molecular
Therapeutics Initiative | UC Berkeley



Rohan Gandhi, PhD
Clinical Translational & BD Strategy |
Autobahn Tx / Vertex



Next 18months. First Prove Demand. Then Upside.

\$5m

Secure Paid Platform Demand



Convert Validation into Recurring Partnership



Create One Partner-Ready Asset Opportunity

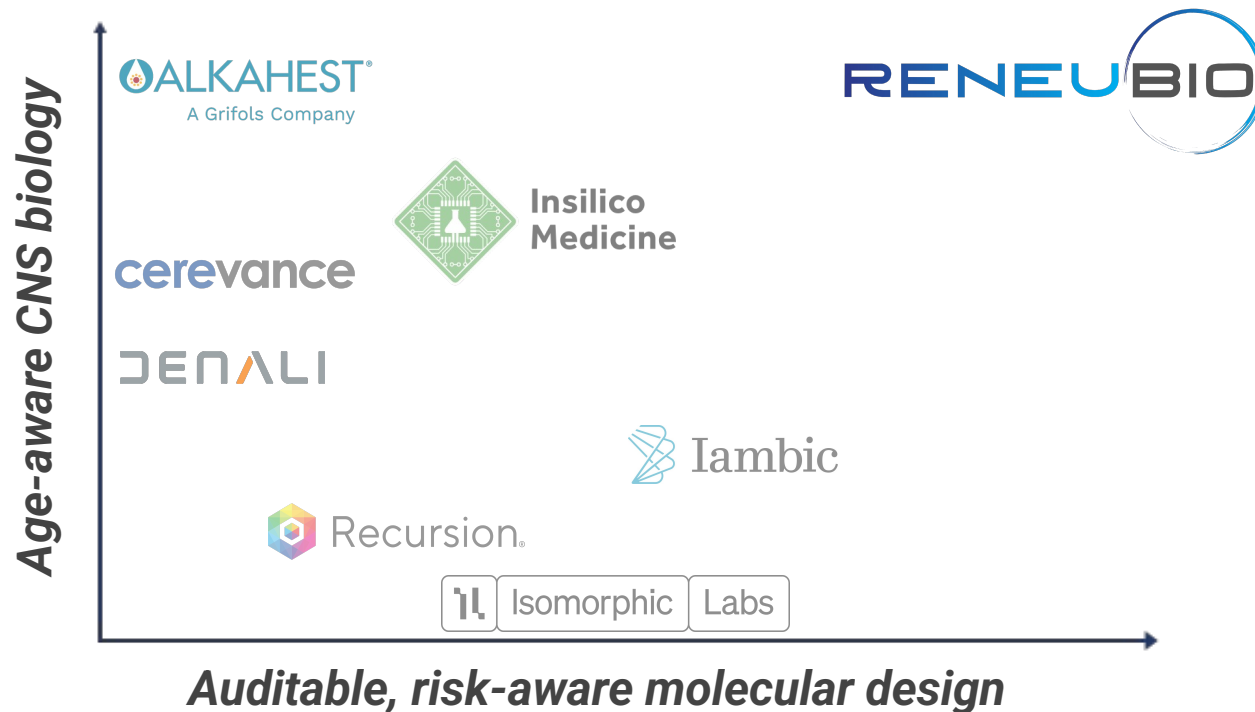
Close 3 paid AlphaForge platform
engagements

Establish 1 recurring pharma/biotech
collaboration

Advance 1 selectively chosen asset to
a partner-ready pre-IND package

Ignore

Auditable, Risk-Aware Design of Partner-Ready Assets



Different Levels of Engagement

Start with high-value decision support. Expand to recurring collaborations.

Single-Asset Readout

\$100K–\$250K / asset

- Risk score + probability by phase
- Failure-mode analysis and scientific rationale
- Written recommendation and redesign options

Portfolio / Diligence Screen

\$250K–\$750K / engagement

- Rank and triage multiple compounds
- Highlight which assets to stop, advance, or redesign
- Best fit for BD, licensing, and portfolio reviews

Recurring Collaboration

\$750K–\$2.0M+ / year

- Ongoing compound evaluation and redesign support
- Regular portfolio reviews with partner teams
- Path to platform licensing and deeper strategic deals

Selective asset upside (secondary): pre-IND partnering, upfront payments, milestones, and downstream royalties where AlphaForge creates a clear advantage.

Pricing is intentionally designed to prove demand first, then expand into recurring and asset-linked economics.

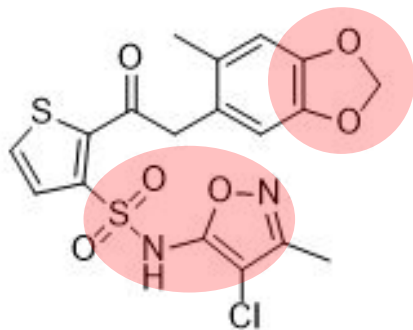
How is AlphaForge different from other AI drug design efforts?

- Drug designs are fully auditable, allowing medchem teams to track the logic
- The model is reverse engineered on clinical trial results, making key clinical risks enforced constraints on the earliest drug designs
 - Our clinical risk assessment algorithm was built to predict probability of clinical failure with an AUC=0.844
- It is the first and only age-aware, brain drug design engine. It leverages internal data, deep expertise in aging brain biology, and evolving as we continue to expand our screening in old human tissue.

Different Levels

- **1. Single-asset diligence readout**
Use for licensing, investment, or internal asset review
Price target: **\$100K–\$250K**
- **2. Multi-asset portfolio screen**
Rank, triage, and identify dominant failure modes across a set of compounds
Price target: **\$250K–\$750K**
- **3. Annual platform collaboration / enterprise partnership**
Repeat access, portfolio support, redesign work, joint reviews
Price target: **\$750K–\$2M+**

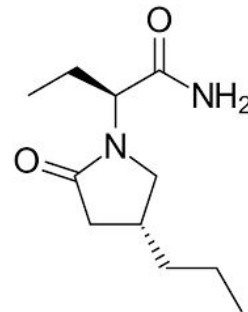
AlphaForge Explains Why a Drug Fails



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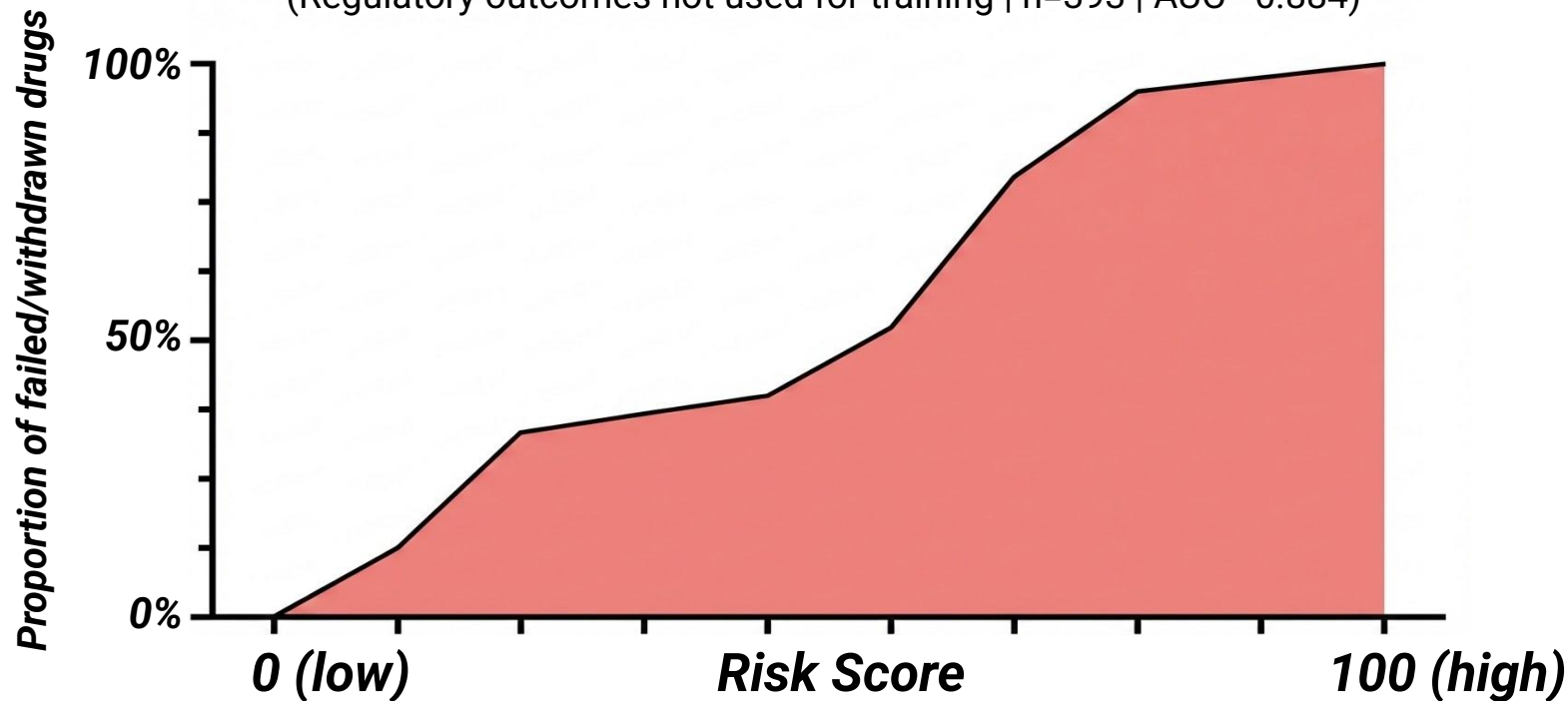
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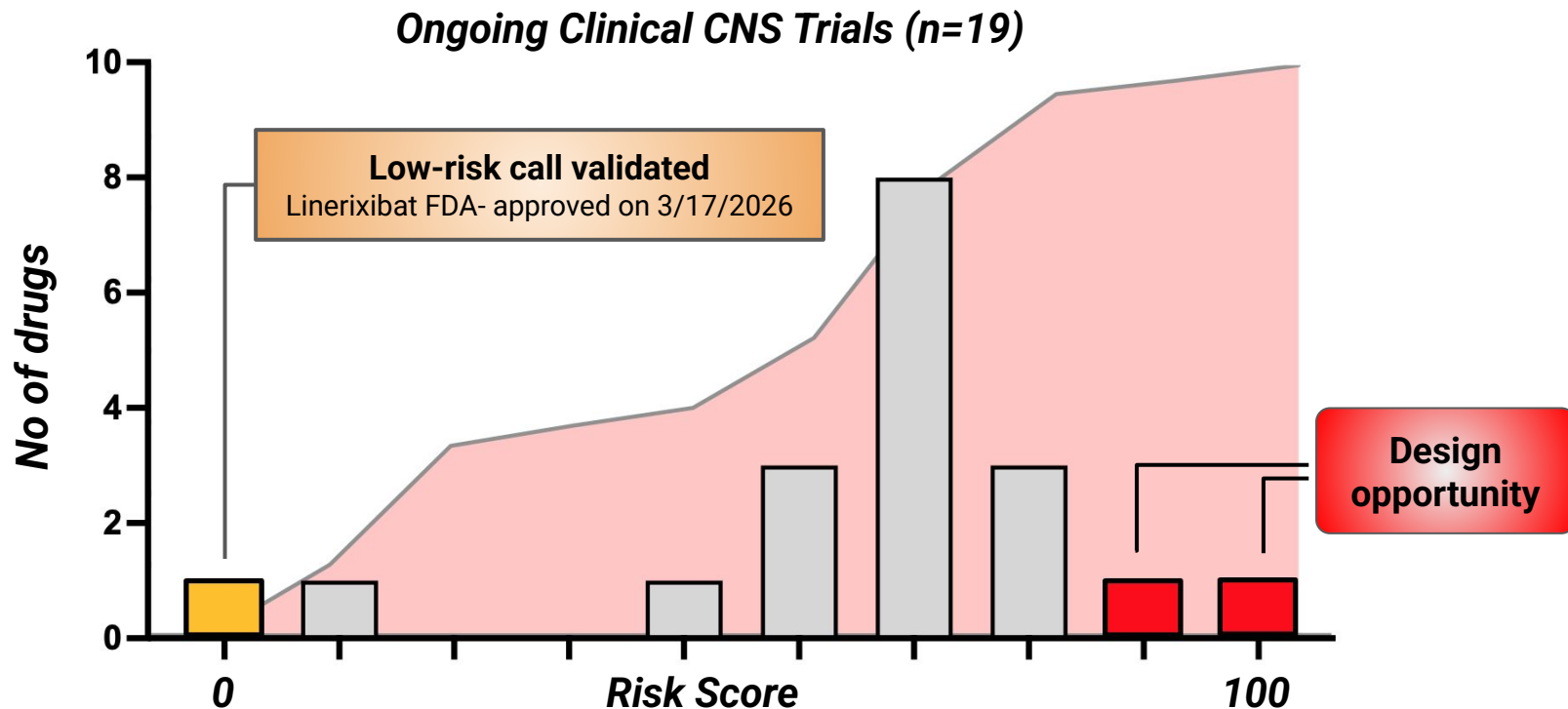
Our Risk Score Tracks Real Clinical Outcomes

Higher score means higher failure risk

(Regulatory outcomes not used for training | n=393 | AUC - 0.884)



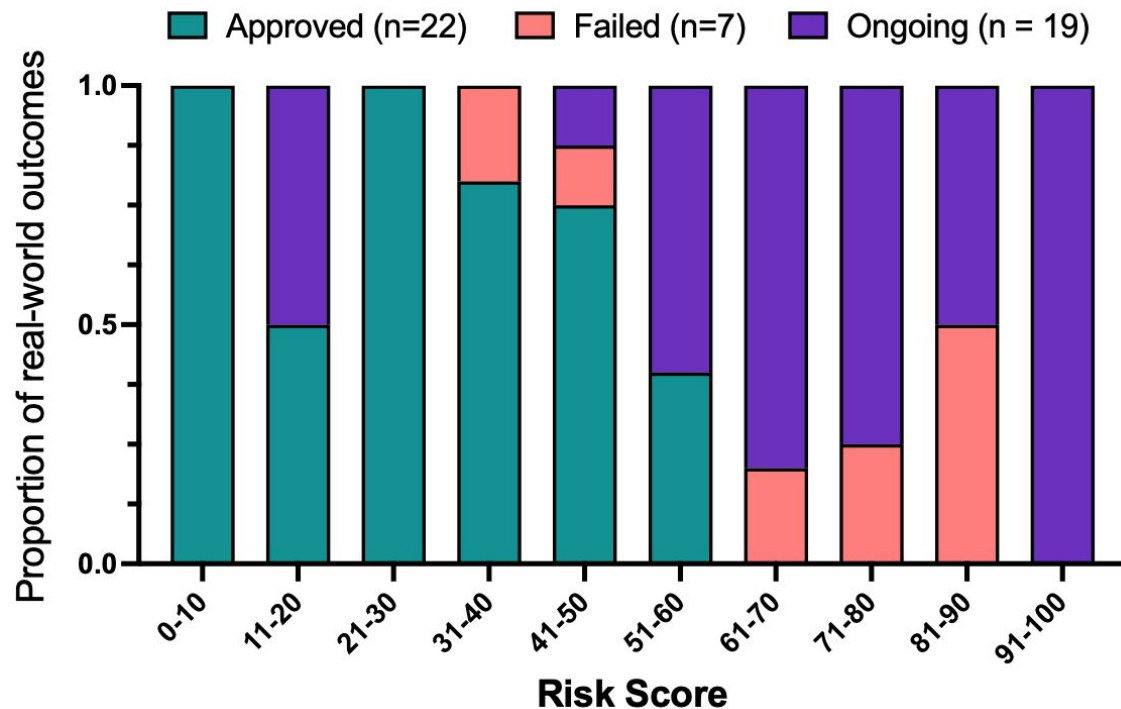
We Are Now Scoring Live Clinical Programs Prospectively



The first low-risk call validated; the rest highlight where redesign may create value

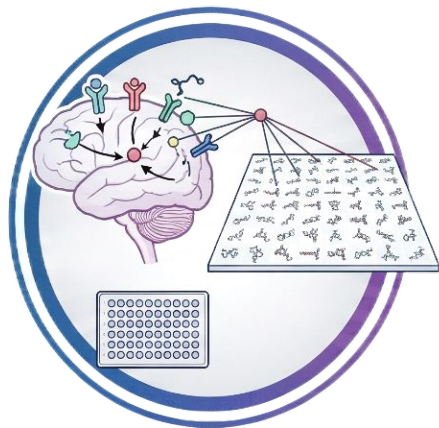
Prospective Evaluation

Drugs with outcomes disclosed after model freeze (2025) or still ongoing



Our Edge Starts with Aging Brain Biology

Aging is the Primary Risk Factor for Most Brain Diseases



- ***Screen in aged brains***
- ***Build predictive models***
- ***Design molecules to treat brain aging & neurodegenerative disease***

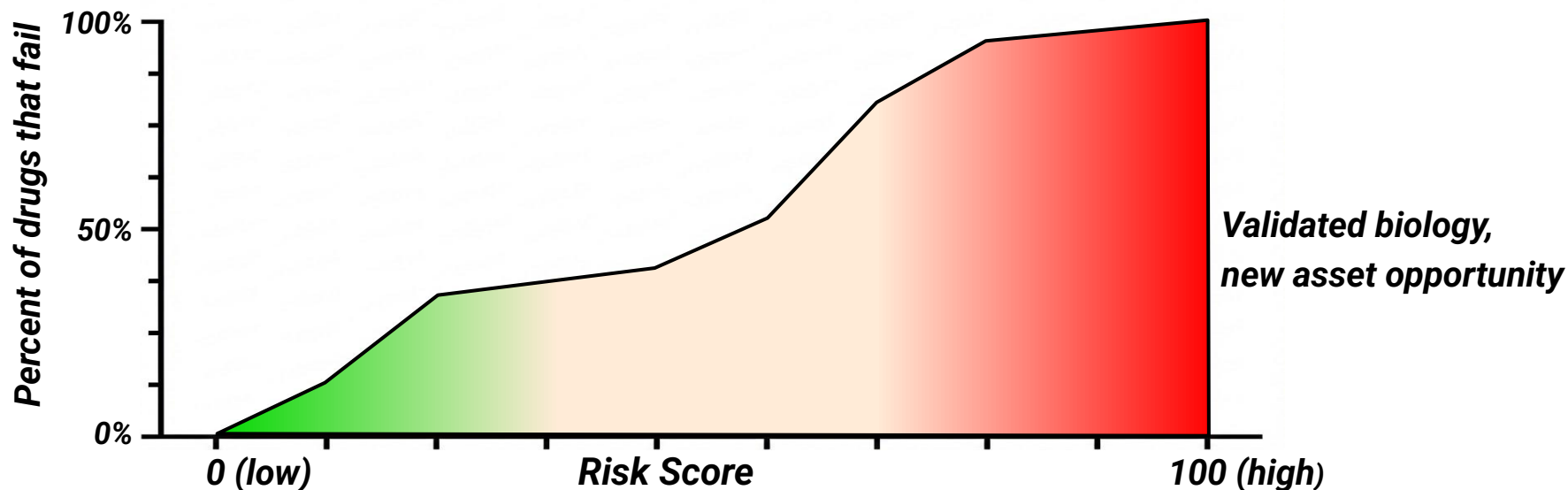
PoC published in **Science** & **Nature** by cofounders:

[Leeman et al. Science, 2018](#); [Ruetz et al. Nature, 2024](#)

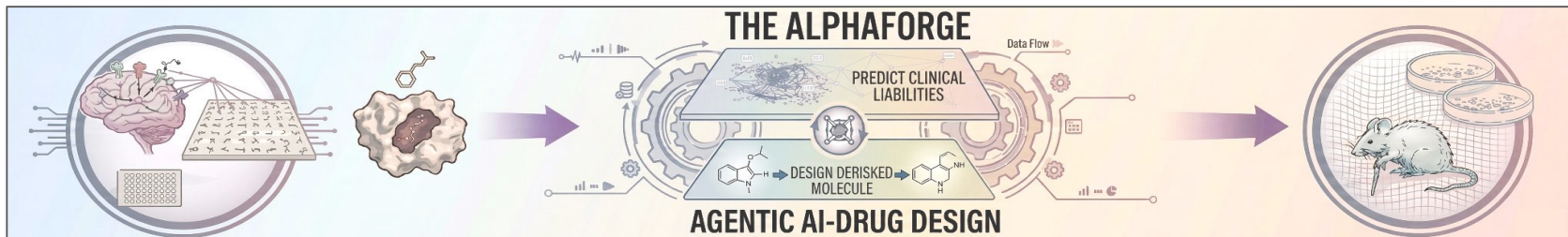
Real Clinical Risk Signal. New Asset Opportunities.

Our risk score tracks real clinical outcomes in CNS.

That lets us find validated biology where better chemistry can create new assets.



Programs 2 & 3: Validated Biology, Better Chemistry

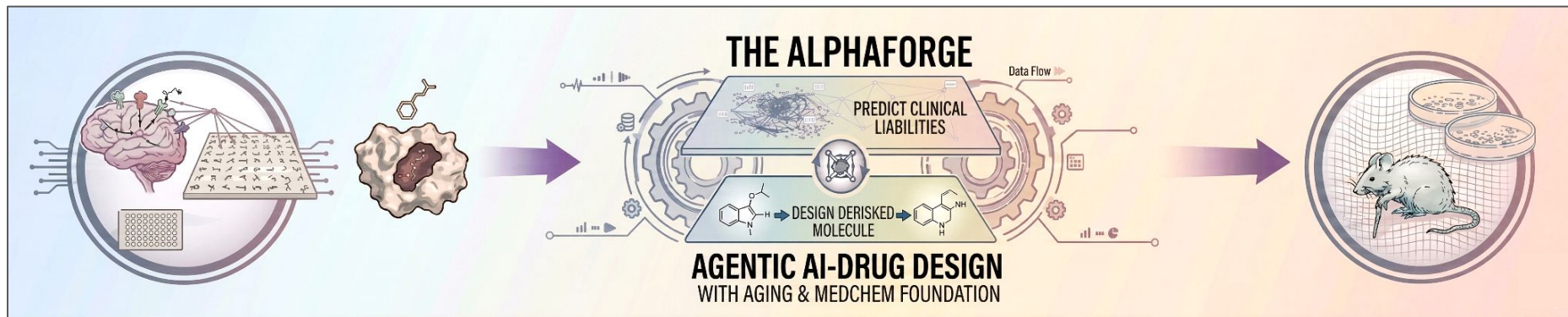


Program	Description	Current Stage	Next Milestone
Validated biology, chemistry constrained	Target validated, chemistry limited	Chemical space mapped / Active redesign ongoing	Designed de-risked leads + early validation
High-value CNS target selected	Commercial relevance + platform expansion	Target selected / active binding-potency modeling	First de-risked leads + early validation

External Pull: UC Berkeley's Molecular Therapeutics Initiative is applying the platform to its programs



Our AI Predicts Failure, Explains Why, and Redesigns Earlier



Score risk

Flag liabilities

Design lower-risk molecules

Supported by 

Developed by co-founders:

Ruetz et al, **Nature**, 2024

Mahmoudi et al, **Nature**, 2019

Leeman, Ruetz,..., Mahmoudi et al, **Science**, 2018



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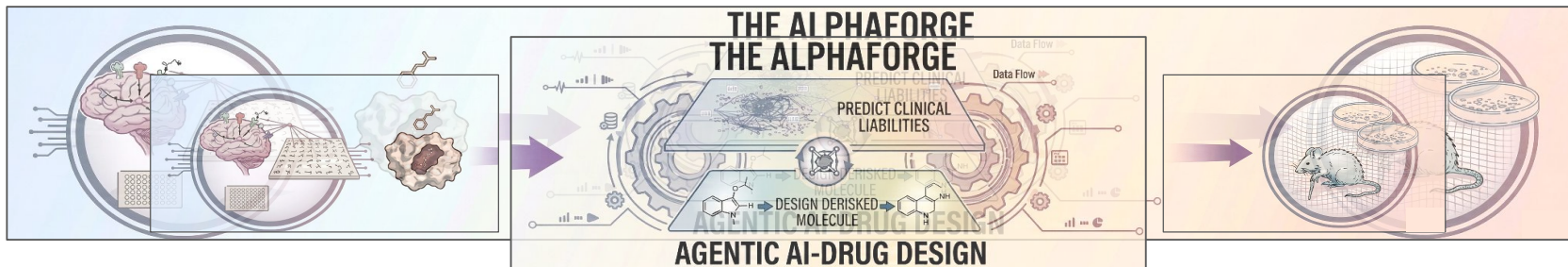
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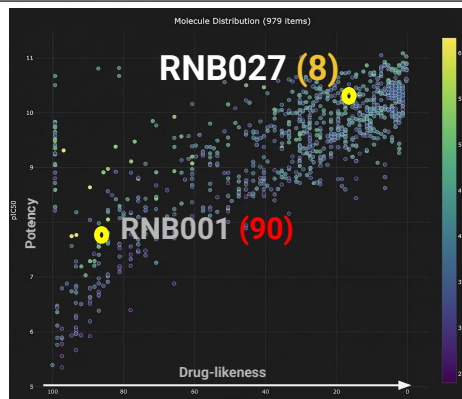
We Used The Platform to Rescue Our Own Program



RNB001 → RNB027

- Reverses brain injury
- ↑ Cognition

Causal regeneration, not symptom control



RNB001 (3rd Gen Academic Lead)

Platform Score: 90 (High Risk)

- ❑ Bis-phenol motif → rapid clearance
- ❑ Redox-active core → toxicity risk

Before

- Scaffold risk score 90 (High Risk)

After

- risk score 8 (Low Risk)
- 3x in vivo PD improvement
- <2 months redesign
- Minimal experimental cost (<\$50K)

Identified by cofounders:

[Ruetz et al. Nature. 2024](#)

Prediction-First Platform Monetization

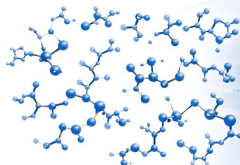
BUILD → DE-RISK → PARTNER

ReneuBio's Platform Generates Partner-Ready CNS Assets



Proprietary AI Platform

Design & optimize small molecules for the aging brain



Owned CNS Asset Engine

Advance to DC-quality / partner-ready preclinical packages

\$4B

Late-stage
(Ph 2)

\$1B

Early clinical /
IND-ready

\$500M

Pre-IND
de-risked assets



Strategic Partnerships

Partner before costly clinical development



Upfront
Payments



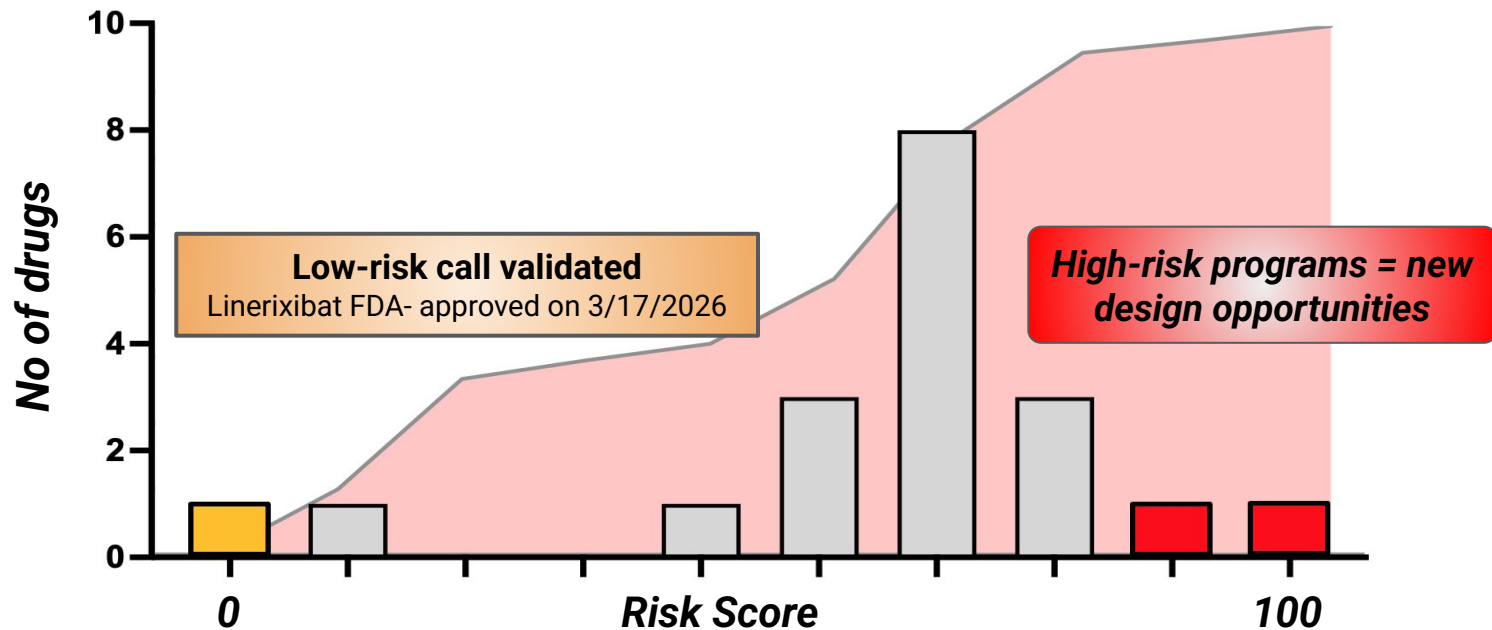
Milestone
Payments



Downstream
Royalties

From Internal Proof to Pipeline Expansion

Ongoing Clinical CNS Trials (n=19)



Ongoing collaboration with UC Berkeley's Molecular  Therapeutics Initiative to apply our platform to their programs

Business Model: Two-Step Commercialization of AlphaForge

- single-asset diligence
- portfolio triage
- liability diagnosis
- redesign recommendations

Step 1 – Primary (short-term priority)

PLATFORM LICENSING & PHARMA PARTNERSHIP

Revenue: paid projects, recurring collaborations, platform licensing

- Single-asset diligence
- Portfolio triage
- Liability diagnosis
- Redesign recommendations

Outcome: near-term revenue, validation, partner penetration, credibility

- Paid pilot projects
- Annual platform collaborations

Value created

- near-term revenue
- external validation
- partner penetration
- credibility and data flywheel

Example economics

- single-asset readout
- multi-asset portfolio screen
- annual collaboration

Step 2 – Secondary (selective upside capture)

INTERNAL ASSETS & SELECTIVE ASSET OPPORTUNITIES

- internal design or external opportunity identification
- use AlphaForge to de-risk and optimize
- monetize through pre-IND partnering or strategic exit

Value created

- higher value capture
- equity-like upside
- controlled capital exposure

Same engine. Sequenced monetization. First prove demand, then selectively own upside.

Business Model: Two-Step Commercialization of AlphaForge

Step 1 — Primary (short-term priority)

PLATFORM LICENSING & PHARMA PARTNERSHIP

AlphaForge is deployed with biotech and pharma partners for:

- single-asset diligence
- portfolio triage
- liability diagnosis
- redesign recommendations

Revenue: paid projects, recurring collaborations, platform licensing

Outcome: near-term revenue, validation, partner penetration, credibility

Step 2 — Secondary (selective upside capture)

INTERNAL ASSETS & SELECTIVE ASSET OPPORTUNITIES

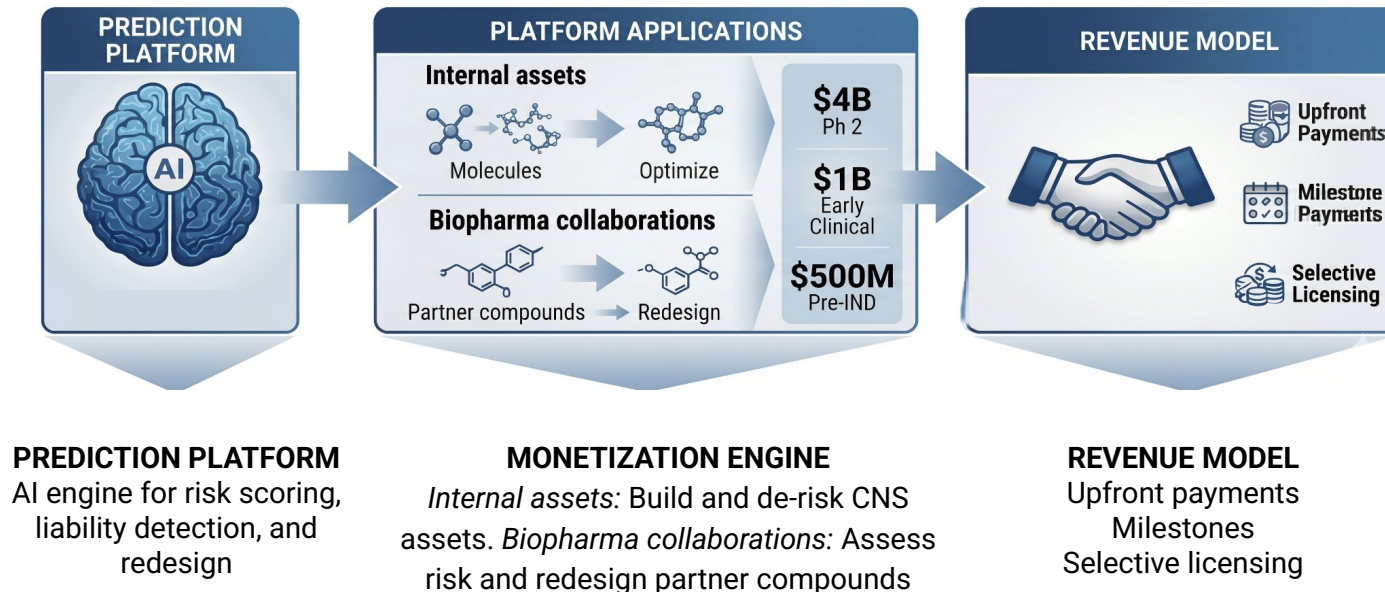
AlphaForge is selectively applied to internal or external assets where de-risking creates outsized value.

Monetization: pre-IND partnering, option deals, selective licensing, strategic exits

Strategic logic: one engine, two sequenced value-capture modes

Same engine. Sequenced monetization. First prove demand, then selectively own upside.

Business Model: Monetize a Prediction-First Drug Design Platform



One Program Forced Us to Build a Platform

- Started as a therapeutics company
- Needed to spot risk earlier
- Built an internal AI prediction engine
- Realized it could scale far beyond one asset



Our AI Predicts Failure, Explains Why, & Redesigns Earlier



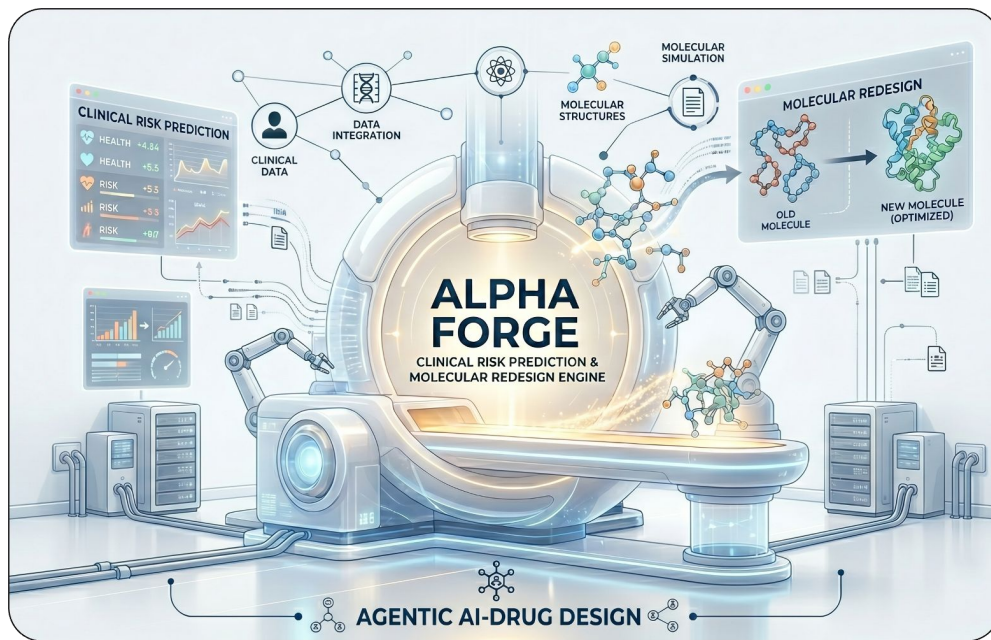
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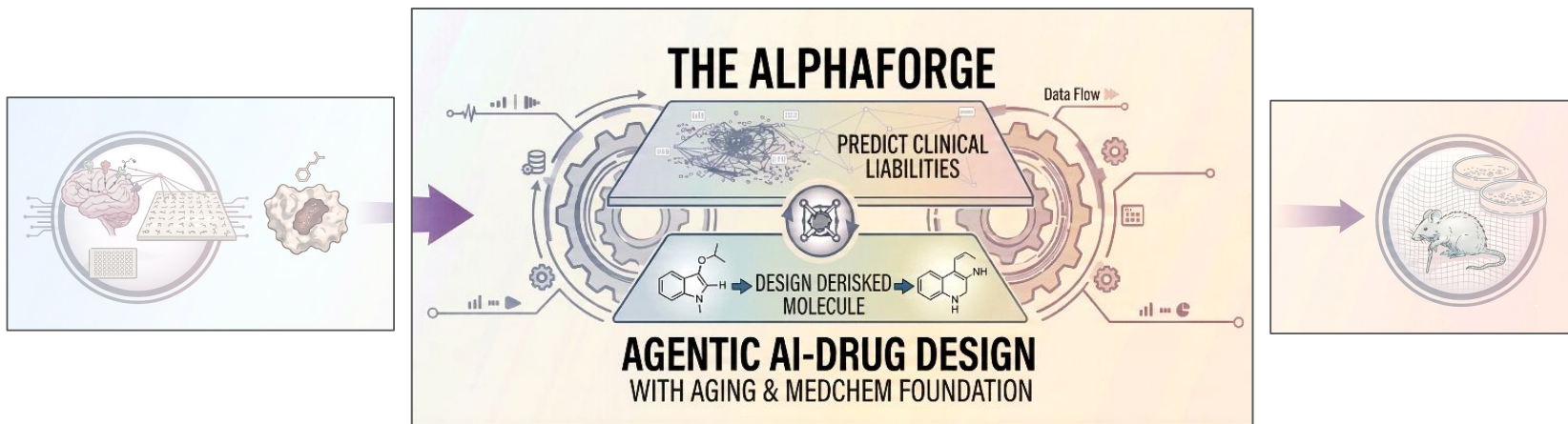
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Supported by **Google**

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This Round Scales the Platform Across Programs

\$5M

Milestones for the Next 18 Months

**Deploy AlphaForge across 3
independent partner- relevant
programs**

**Secure 2 paid commercial
engagements**

**Build a repeatable
partner-delivery engine**